Title = E-Commerce Delivery Analytics

Project Overview

This project performs Exploratory Data Analysis (EDA) on an e-commerce delivery dataset to understand factors affecting delivery performance, customer satisfaction, refund requests, and order value patterns.

The analysis uses Python to explore delivery delays, platform performance, and product category trends to generate insights that could help improve operational efficiency and customer experience.

Business Questions

This analysis aims to answer the following key business questions:

1) Which delivery platform processes the highest number of transactions?

2) Is delivery delay associated with a higher probability of refund requests?

3) Which product categories experience the highest number of refund requests?

4) Which product categories generate the highest total order value?

5) Which operational factors (such as delivery time or delay) have the greatest impact on customer service ratings?

6) Which delivery platform achieves the highest average customer service ratings?

Importing Libraries

In [1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as pt
import seaborn as sn
```

Exploratory Data Analysis

In [2]:
```
#Reading Data
data = pd.read_csv('Ecommerce_Delivery_Analytics_New.csv')
```

In [3]:
```
#Exploring data
data.head(10) # We have 100000 Rows and 11 Columns
```

Out[3]:

	Order ID	Customer ID	Platform	Order Date & Time	Delivery Time (Minutes)	Product Category	Order Value (INR)	Customer Feedback	Service Rating	Delivery Delay	Refund Requested
0	ORD000001	CUST2824	JioMart	19:29.5	30	Fruits & Vegetables	382	Fast delivery, great service!	5	No	No
1	ORD000002	CUST1409	Blinkit	54:29.5	16	Dairy	279	Quick and reliable!	5	No	No
2	ORD000003	CUST5506	JioMart	21:29.5	25	Beverages	599	Items missing from order.	2	No	Yes
3	ORD000004	CUST5012	JioMart	19:29.5	42	Beverages	946	Items missing from order.	2	Yes	Yes
4	ORD000005	CUST4657	Blinkit	49:29.5	30	Beverages	334	Fast delivery, great service!	5	No	No
5	ORD000006	CUST3286	Blinkit	36:29.5	25	Personal Care	1939	Items missing from order.	2	No	Yes
6	ORD000007	CUST2679	JioMart	22:29.5	22	Personal Care	1693	Fast delivery, great service!	5	No	No
7	ORD000008	CUST9935	Swiggy Instamart	50:29.5	16	Personal Care	195	Horrible experience, never ordering again.	1	No	Yes
8	ORD000009	CUST2424	JioMart	51:29.5	39	Fruits & Vegetables	182	Very satisfied with the service.	5	No	No
9	ORD000010	CUST7912	JioMart	08:29.5	51	Grocery	1279	Very late delivery, not happy.	2	Yes	Yes

In [4]:
```
data.tail()
```


Out[4]:

	Order ID	Customer ID	Platform	Order Date & Time	Delivery Time (Minutes)	Product Category	Order Value (INR)	Customer Feedback	Service Rating	Delivery Delay	Refund Requested
99995	ORD099996	CUST5324	JioMart	49:29.5	24	Dairy	289	Packaging could be better.	3	No	No
99996	ORD099997	CUST1677	JioMart	18:29.5	19	Snacks	322	Good quality products.	4	No	No
99997	ORD099998	CUST8198	JioMart	27:29.5	41	Dairy	135	Fast delivery, great service!	5	Yes	No
99998	ORD099999	CUST9975	JioMart	14:29.5	31	Grocery	973	Quick and reliable!	5	No	No
99999	ORD100000	CUST3748	JioMart	41:29.5	34	Fruits & Vegetables	453	Packaging could be better.	3	No	No

In [5]:
```
#statistical Exploration of Data
data.describe()
```

Out[5]:

	Delivery Time (Minutes)	Order Value (INR)	Service Rating
count	100000.000000	100000.000000	100000.000000
mean	29.536140	590.994400	3.240790
std	9.958933	417.409058	1.575962
min	5.000000	50.000000	1.000000
25%	23.000000	283.000000	2.000000
50%	30.000000	481.000000	3.000000
75%	36.000000	770.000000	5.000000
max	76.000000	2000.000000	5.000000

In [6]:
```
data.info()
#No nulls Values in the data . So Data Cleaning not needed .
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 100000 entries, 0 to 99999
Data columns (total 11 columns):
 #   Column                   Non-Null Count   Dtype
---  ------                   --------------   -----
 0   Order ID                 100000 non-null  object
 1   Customer ID              100000 non-null  object
 2   Platform                 100000 non-null  object
 3   Order Date & Time        100000 non-null  object
 4   Delivery Time (Minutes)  100000 non-null  int64
 5   Product Category         100000 non-null  object
 6   Order Value (INR)        100000 non-null  int64
 7   Customer Feedback        100000 non-null  object
 8   Service Rating           100000 non-null  int64
 9   Delivery Delay           100000 non-null  object
 10  Refund Requested         100000 non-null  object
dtypes: int64(3), object(8)
memory usage: 8.4+ MB
```

In [7]:
```
data.isnull().sum() # No Null Values
```


Out[7]:
```
Order ID                   0
Customer ID                0
Platform                   0
Order Date & Time          0
Delivery Time (Minutes)    0
Product Category           0
Order Value (INR)          0
Customer Feedback          0
Service Rating             0
Delivery Delay             0
Refund Requested           0
dtype: int64
```

In [8]:
```
# Renaming Column
data= data.rename(columns = {'Delivery Time (Minutes)':'Delivery Time'})
```

In [9]:
```
data.head()
```

Out[9]:

	Order ID	Customer ID	Platform	Order Date & Time	Delivery Time	Product Category	Order Value (INR)	Customer Feedback	Service Rating	Delivery Delay	Refund Requested
0	ORD000001	CUST2824	JioMart	19:29.5	30	Fruits & Vegetables	382	Fast delivery, great service!	5	No	No
1	ORD000002	CUST1409	Blinkit	54:29.5	16	Dairy	279	Quick and reliable!	5	No	No
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3	ORD000004	CUST5012	JioMart	19:29.5	42	Beverages	946	Items missing from order.	2	Yes	Yes
4	ORD000005	CUST4657	Blinkit	49:29.5	30	Beverages	334	Fast delivery, great service!	5	No	No

Problem Statement Analysis

Question 1 - Which delivery platform processes the highest number of transactions?

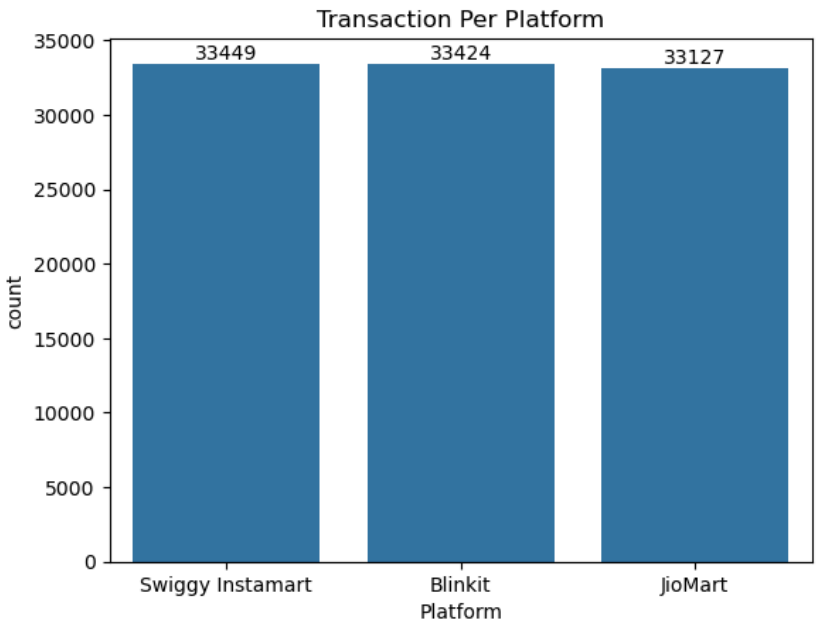
Analysis

In [12]:
```
count=data.groupby('Platform')['Service Rating'].count().sort_values(ascending = False)
count
```

Out[12]:
```
Platform
Swiggy Instamart    33449
Blinkit             33424
JioMart             33127
Name: Service Rating, dtype: int64
```

Visualization

In [17]:
```
ax = sn.countplot(x="Platform", data=data,order = count.index)
for container in ax.containers: # Adding labels
    ax.bar_label(container)
pt.title("Transaction Per Platform")
pt.show()
```

Observation

Swiggy Instamart processes the highest number of transactions compared to other platforms.

Insight

Swiggy Instamart likely has greater customer reach or operational efficiency compared to competitors.

Business Recommendation

The company could analyze the operational and marketing strategies of Swiggy Instamart to improve performance across other platforms.

In [ ]:

Question 2 - Is delivery delay associated with higher refund probability?

Visualization

In [14]:
```
sn.boxplot(x='Refund Requested',y ='Delivery Time',data=data)
pt.title('Delivery vs Refund Requests')
pt.show()
```

Observation

Orders that requested refunds has almost same delivery time as for those which has not requested refunds

Insight

Delivery delays appear to have no or very less contribution in customer dissatisfaction and refund requests.

Recommendation

Since delivery time does not significantly influence refund requests, the company should focus on product quality, service experience, and category level issues to reduce refund rates.

In [ ]:

```
```

Question 3 - Which product category gets most refund request ?

Analysis

In [15]:

```
refund_requested = data[data['Refund Requested'] == 'Yes']
sort = refund_requested['Product Category'].value_counts().sort_values(ascending = False) # Giving count of refunds of each Pr
sort
```

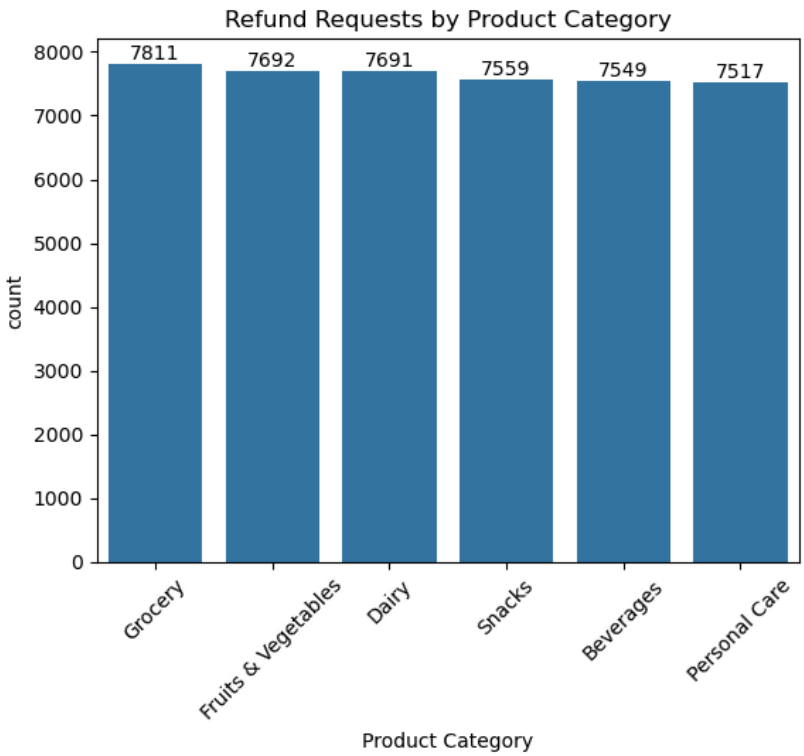
Out[15]:

```
Product Category
Grocery                7811
Fruits & Vegetables    7692
Dairy                  7691
Snacks                 7559
Beverages              7549
Personal Care          7517
Name: count, dtype: int64
```

Visualization

In [18]:

```
a = sn.countplot(x = 'Product Category',data = refund_requested,order = sort.index)
for container in a.containers:
    a.bar_label(container)
pt.title('Refund Requests by Product Category')
pt.xticks(rotation=45) #Rotating x axis Values to 45 For clear Identification
pt.show()
```

Observation

Grocery has the highest amount of refund request.

Insight

Grocery orders may be more prone to quality or delivery-related issues.

Recommendation

Improve quality checks and packaging for perishable items like grocery , Fruits , Dairy orders to reduce refunds.

In [ ]:

Question 4 - Which Products Category generate Highest Order Value ?

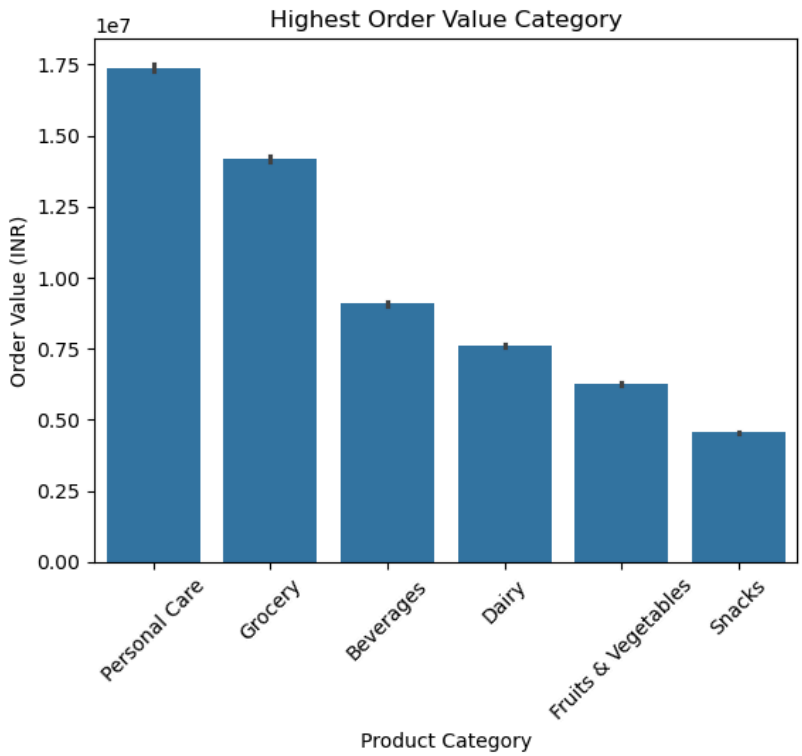
Analysis

In [19]:
```
order=data.groupby('Product Category')['Order Value (INR)'].sum().sort_values(ascending = False)
order
```

Out[19]:
```
Product Category
Personal Care          17395601
Grocery                14194055
Beverages               9086669
Dairy                   7610522
Fruits & Vegetables     6246517
Snacks                  4566076
Name: Order Value (INR), dtype: int64
```

Visualization

In [20]:
```
ax=sn.barplot(x='Product Category',y='Order Value (INR)',data = data, estimator=sum,order=order.index)
pt.title('Highest Order Value Category')
pt.xticks(rotation=45)
pt.show()
```

Observation

Personal Care generates the highest total order value, followed by Grocery and Beverages.

Insight

Customers spend more on Personal Care and Grocery products, making them the primary value driving categories.

Business Recommendation

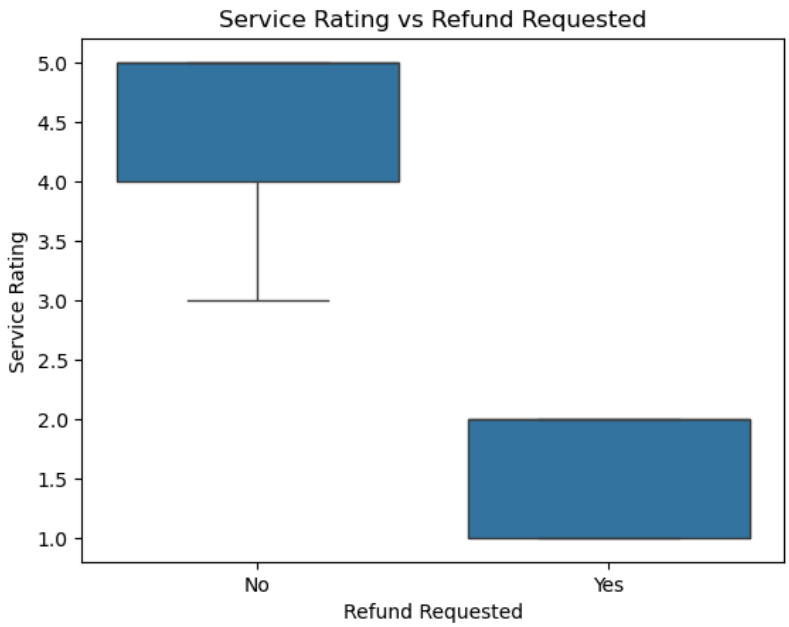
Focus promotions and inventory optimization on high value categories to maximize sales potential.

In [ ]:

Question 5 - Which Factor Impacts Customer Rating the Most?

Visualization

In [24]:
```
sn.boxplot(x='Refund Requested',y='Service Rating',data = data)
pt.title("Service Rating vs Refund Requested")
pt.show()
```

Observation

Orders with refund requests tend to receive significantly lower service ratings.

Insight

Refund requests are strongly associated with lower customer satisfaction.

Business Recommendation

Reducing operational issues that lead to refunds could improve overall customer ratings.

In [ ]:

Question 6 - Which delivery platform achieves the highest average customer service ratings?¶

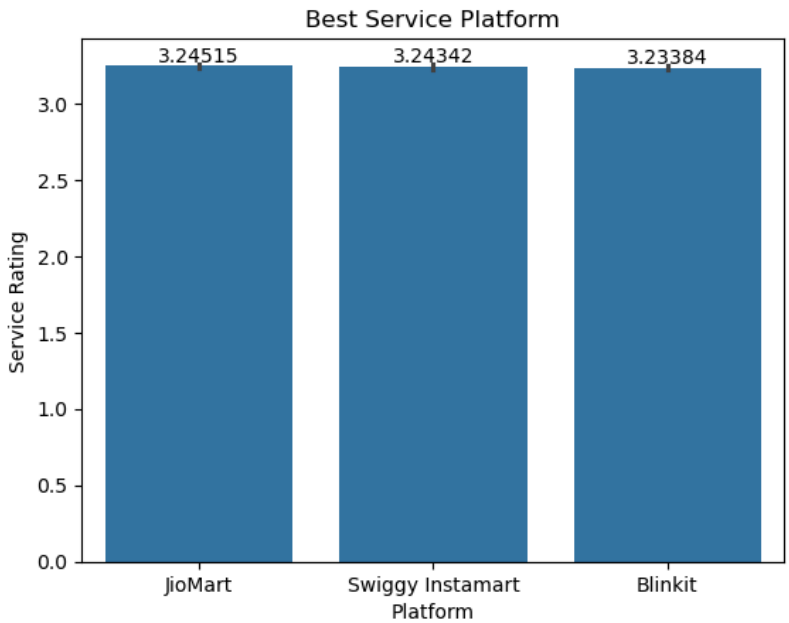
Analysis

In [26]:
```
rating = data.groupby('Platform')['Service Rating'].mean().sort_values(ascending = False)
rating
```

Out[26]:
```
Platform
JioMart             3.245147
Swiggy Instamart    3.243415
Blinkit             3.233844
Name: Service Rating, dtype: float64
```

Visualization

In [28]:
```
ax = sn.barplot(x='Platform',y = 'Service Rating',data = data , order = rating.index)
for container in ax.containers:
    ax.bar_label(container)
pt.title('Best Service Platform')
pt.show()
```

Observation

JioMart has the highest average ratings, followed closely by Swiggy Instamart and Blinkit.

Insight

JioMart appears to deliver slightly better customer service experiences.

Business Recommendation

Analyze JioMart operational practices to identify factors contributing to higher customer satisfaction.

In [ ]:

Key Insights

Personal Care and Grocery categories generate the highest total order value, indicating stronger customer spending in these segments.

Grocery category records the highest number of refund requests, suggesting potential operational or product quality issues.

Orders associated with refunds show significantly lower service ratings, indicating a strong link between refund cases and customer dissatisfaction.

JioMart appears to have best average delivery ratings, although performance across platforms remains relatively close.

In [ ]: